

PROIECTAREA CU MICROPROCESOARE

C3 2025-2026 Sem. 2

Proiect 5

Sapt 5

25 martie 2026

19:00-20:00

serban@upit.ro

Regulament disciplina

Nota finala este formata din activitatile:

- Laborator 15%
- Lucrare control 15%
- Prezentare si sustinere proiect 20%
- Examen 50%
- Bonus prezenta – 10%

Conditii pentru promovare:

- Nota de la laborator trebuie sa fie minim 5 (prezenta obligatorie la toate sedintele de laborator);
- Nota proiect minim 5 (conditie obligatorie pentru intrare in examen);
- Nota de la examen trebuie sa fie minim 5;
- Nota de minim 5 la Lucrarea de control.

In cazul reluarii disciplinei intr-un alt an universitar, activitatile nepromovate trebuie parcurse din nou.

Cerinte proiect

1). Realizarea suportului scris (format pdf) care sa contina capitolele:

Proiectarea Hardware si Proiectarea Software

- In cap. Proiectarea Hardware trebuie sa fie inclusa argumentarea pentru logica de conectare a circuitelor de memorie si I/O, respectiv a perifericelor;
- Este **obligatorie** schema de conectare finala a tuturor componentelor (eventual fara circuitele de alimentare) cu precizarea tuturor pinilor circuitelor folosite si a modului de conectare a acestora; va fi prezentat modul de programare a tablei prin folosirea tastelor (manualul de utilizare);
- In cap. Proiectarea Software trebuie sa fie inclusa logica de constructie a programului software cu toate componentele sale (prin descrieri explicative si organigrame), respectiv listingul programului;
- Este **obligatorie** lista de variabile folosite sub forma unui tabel de forma:

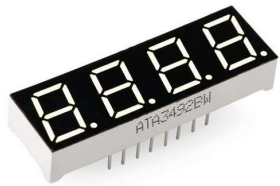
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM

- Folosirea unui simulator pentru programul (sau portiuni din el) realizat este un bonus.

2). Predarea proiectului se va face pana in ziua de **23 mai 2026 ora 23:00** pe platforma moddle.

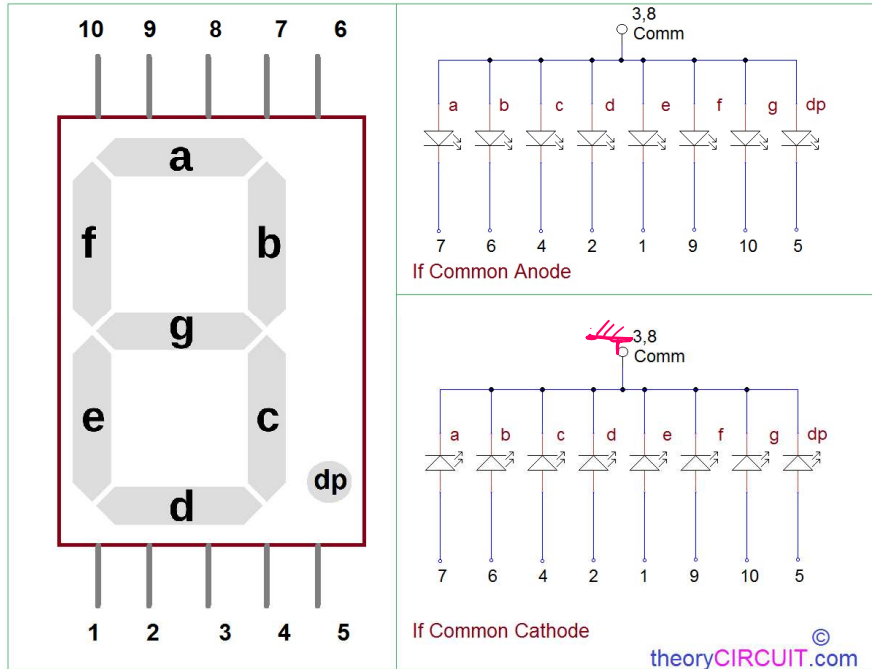
3). Sustinerea proiectului de catre realizator (activitate obligatorie) printr-o sesiune de intrebari legate de solutiile folosite (in ultima saptamana din semestru 25 – 29 mai 2026 la o data stablita de comun acord).

Celule de afisaj LED cu 7 segmente



Celule de afisaj LED cu 7 segmente

7 Segment Display Pinout

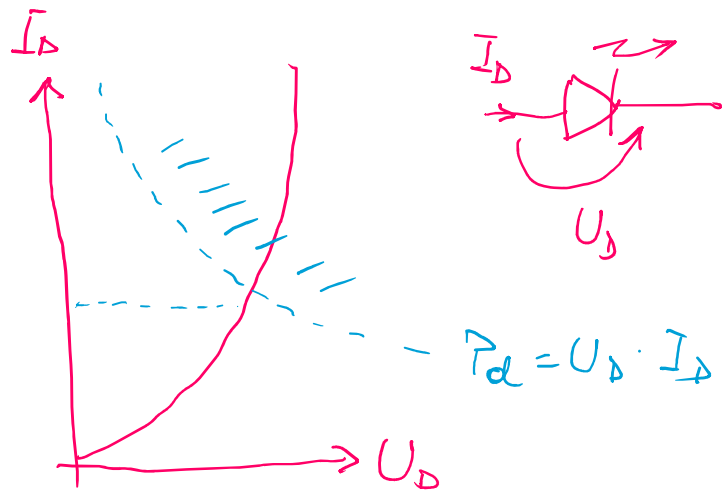
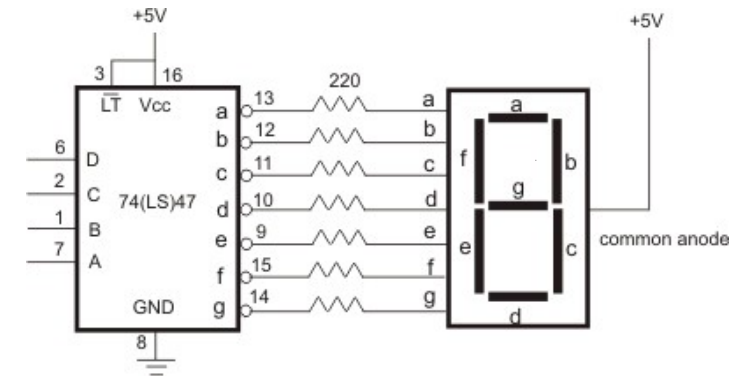
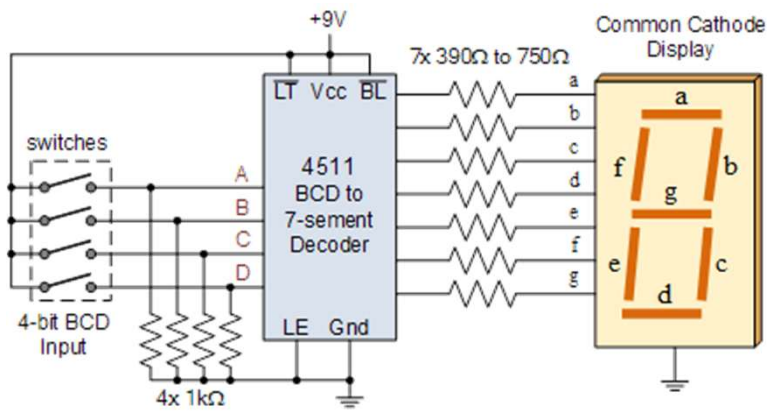


CATHODE COMMON

segment outputs							display
a	b	c	d	e	f	g	
1	1	1	1	1	1	0	0
0	1	1	0	0	0	0	1
1	1	0	1	1	0	1	1
1	1	1	1	0	0	1	1
0	1	1	0	0	1	1	1
1	0	1	1	0	1	1	1
0	0	1	1	1	1	1	1
1	1	1	0	0	0	0	1
1	1	1	1	1	1	1	1
1	1	1	0	0	1	1	1

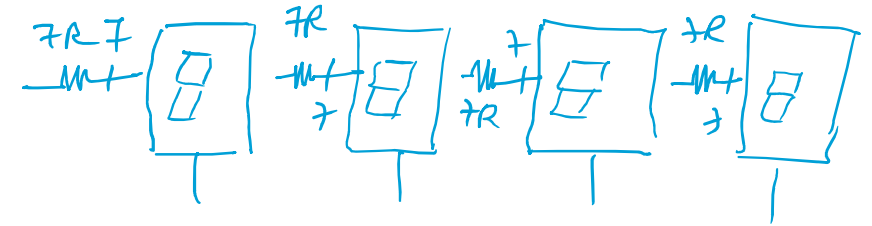
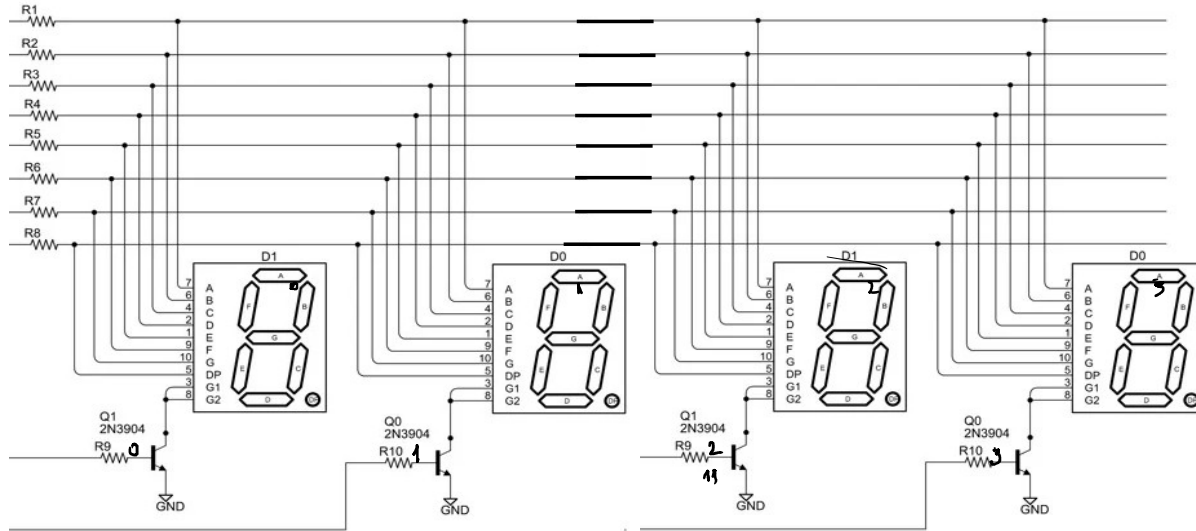
7-Segment Display

Celule de afisaj LED cu 7 segmente



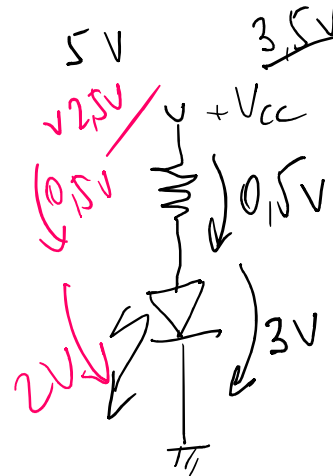
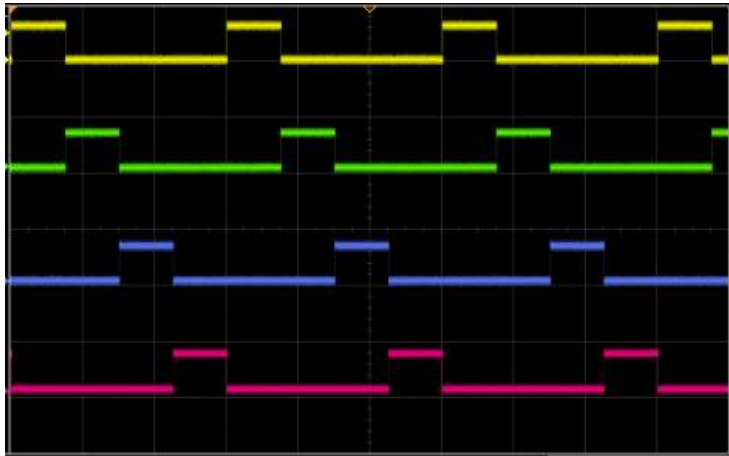
comanda clasic pentru
celula LED 7 segmente -

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



$7R \times 4 = 28R$ Rezistente
 $7 \times 4 = 28$ fire comanda
 4 - catod -
 (comanda clasic)

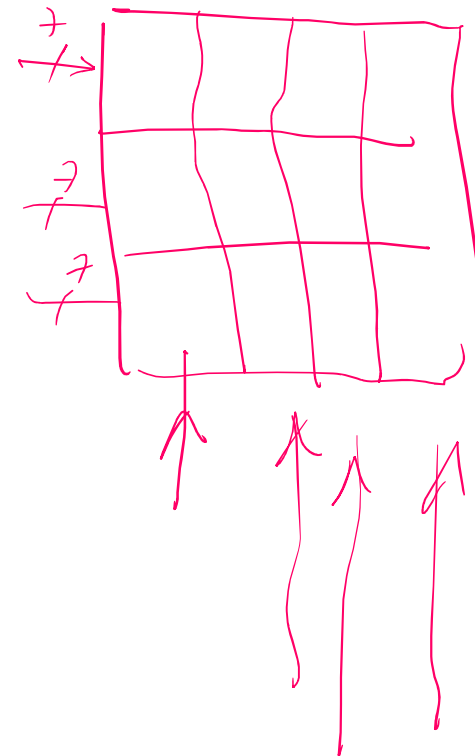
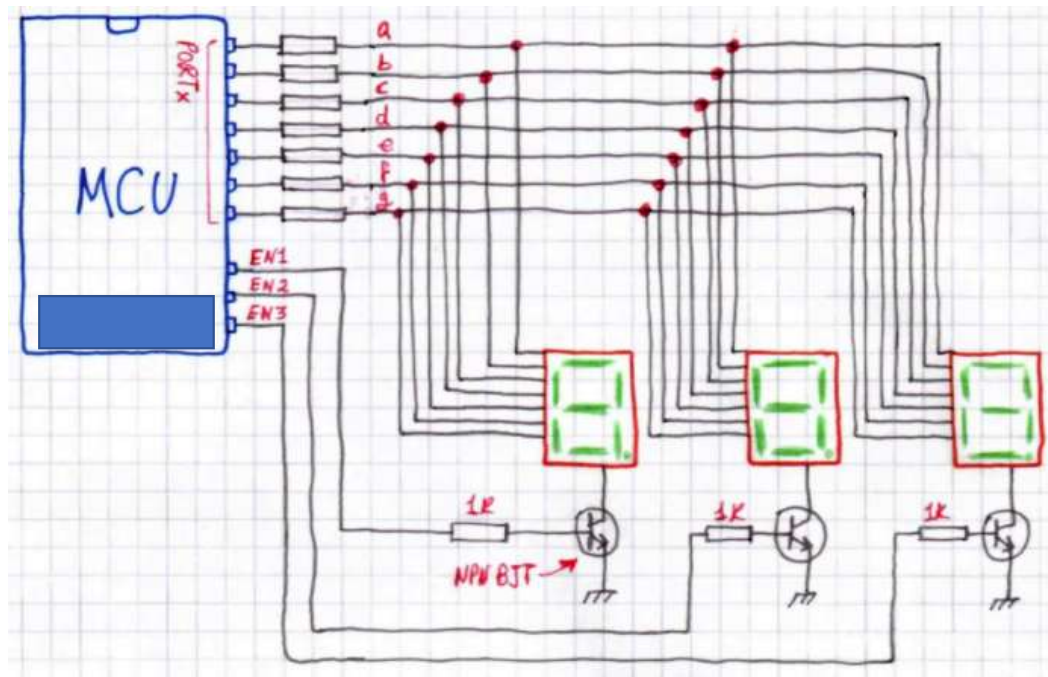
$t = ?$
 $f > 50 \text{ Hz}$ $T < 20 \text{ ms}$

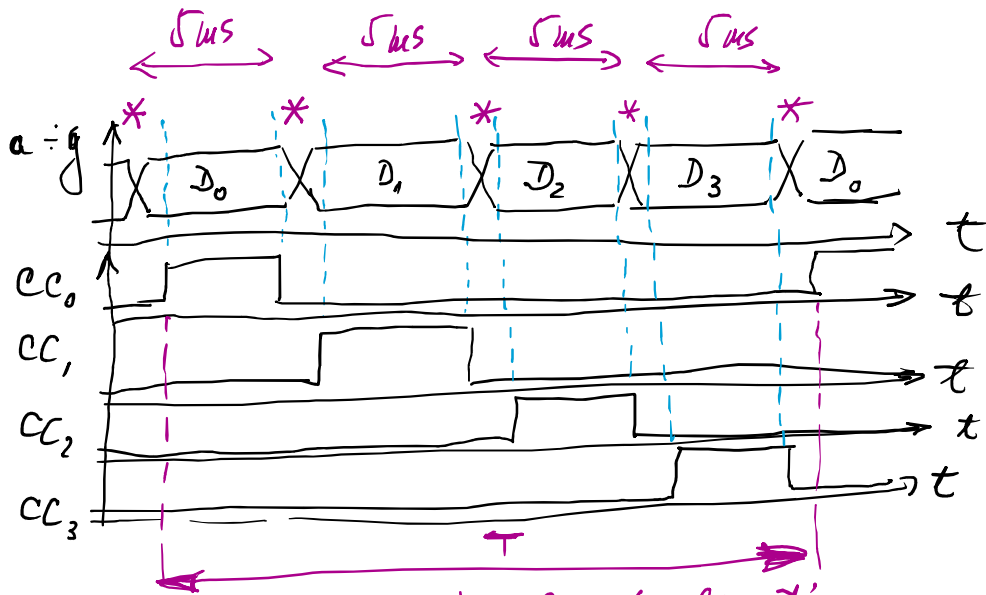


Comanda prin multiplexare
 Resurse

- $7R$ - rezistente
- $7 + 4 = 11$ fire comanda

Proiectarea afisajului cu celule LED 7 segmente comandat prin multiplexare



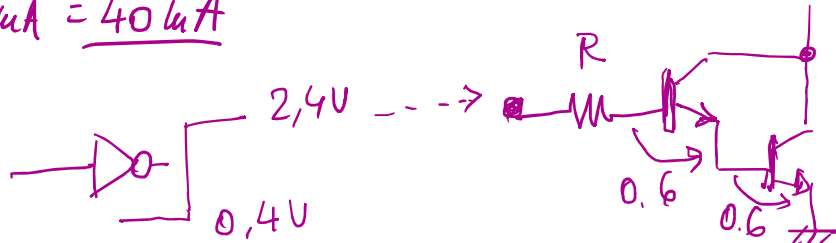


$T \leq 20 \text{ ms}$ ($f \geq 50 \text{ Hz}$) $n = 4$ digits
 $T/n = \frac{20 \text{ ms}}{4} = 5 \text{ ms} \Rightarrow$ necesitate folosire timer

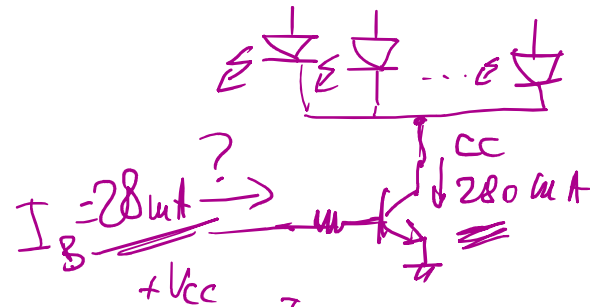
$\Rightarrow$ semnalizari la 5 ms prin cereri de intrerupere catre CPU

$$I_{\text{peak}} \geq n \cdot I_D = 4 \cdot 10 \text{ mA} = 40 \text{ mA}$$

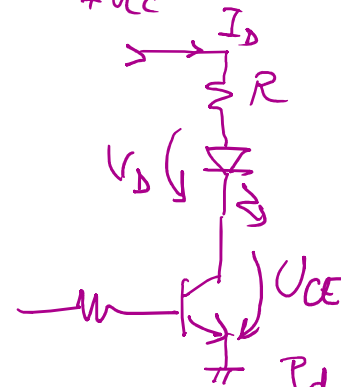
$\Rightarrow$ necesitate circuite amplificare curent



Circuitul de amplificare in curent α_2 (cotelod comun) -



medie putere BD 141, 143,



$$V_{CC} = I_D \cdot R + V_D + U_{CE}$$

$$5 = 40 \cdot R + 2 + U_{CE \text{ SAT}}$$

$$R = \frac{5 - 2 - 1}{40} = \frac{2}{40} = 0,05 \text{ k}$$

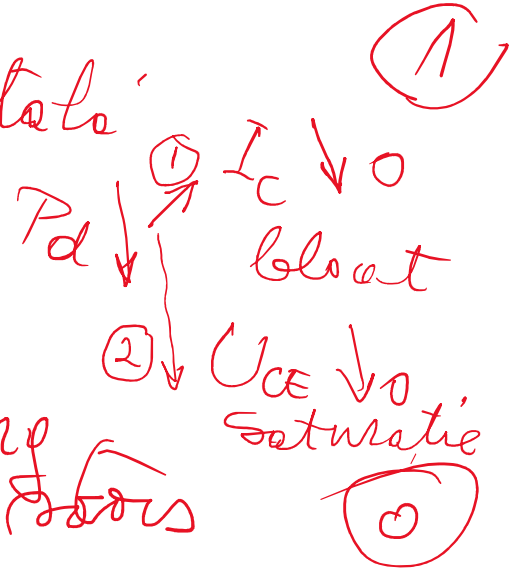
$$\beta_{\text{SAT}} = 10!$$

Regim saturat $U_{CE} \rightarrow 0$
 Regim blocat $I_C \rightarrow 0$

Operare cu tranzistori în logica digitală

1) Putere disipată

$$P_{d\text{tranz}} = U_{CE} \cdot I_C$$



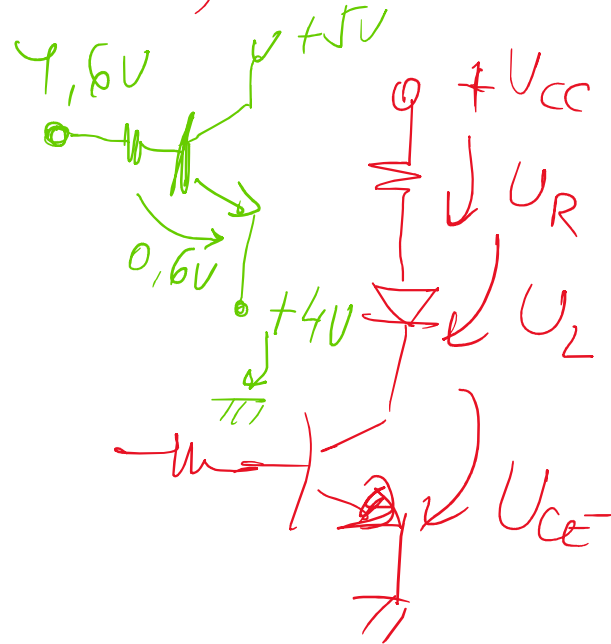
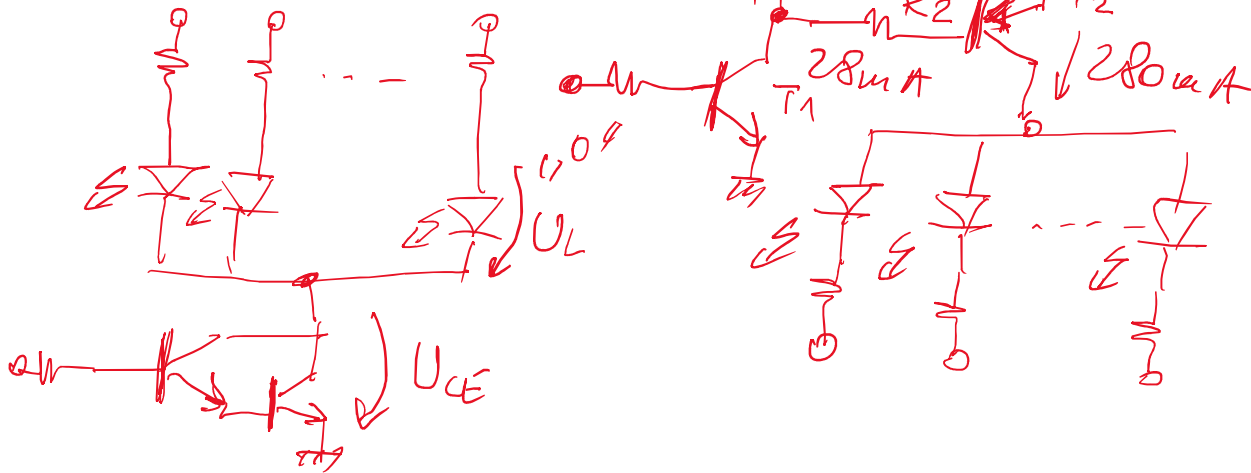
Switching transistors

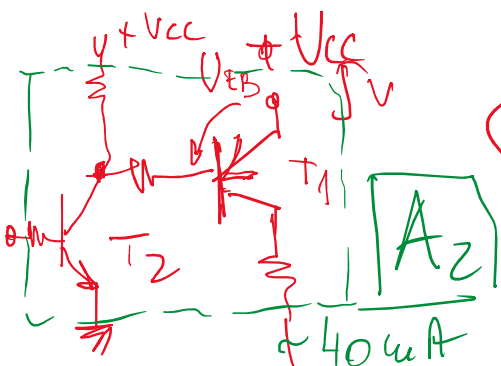
$$P = U \cdot I = I \cdot R \cdot I = I^2 \cdot R$$



Catod comun

Anod comun

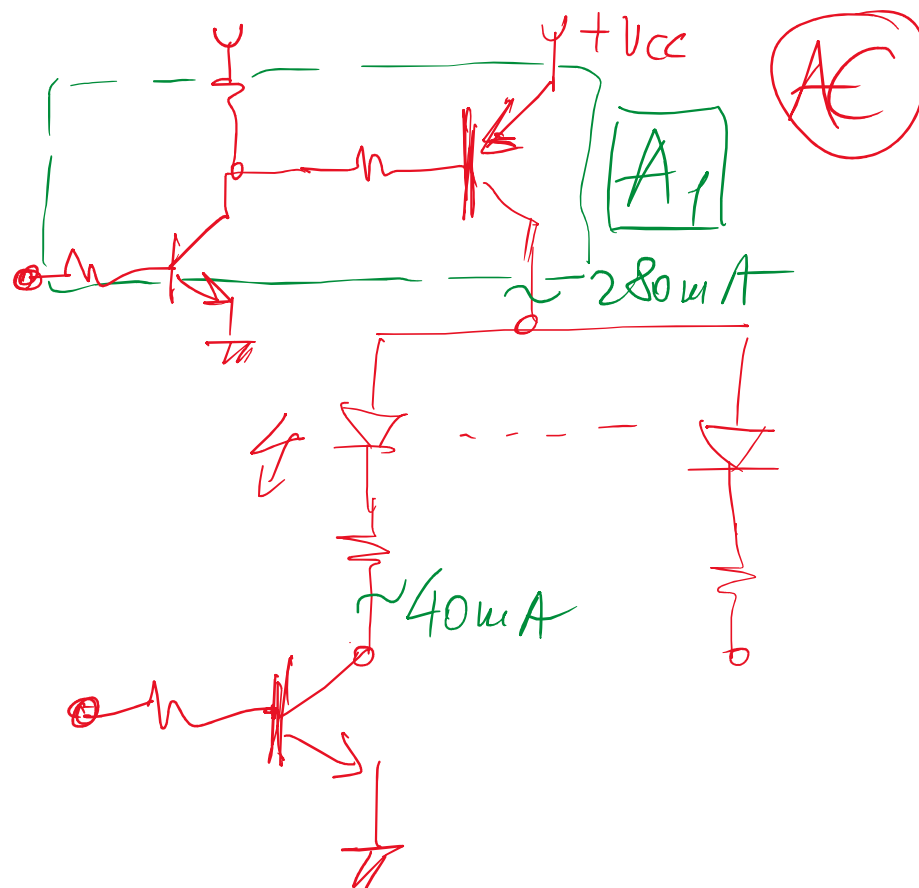
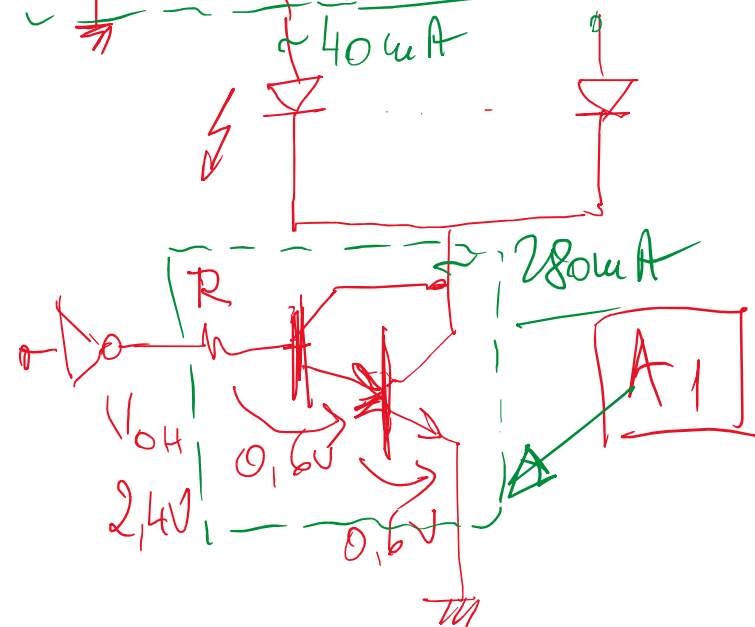




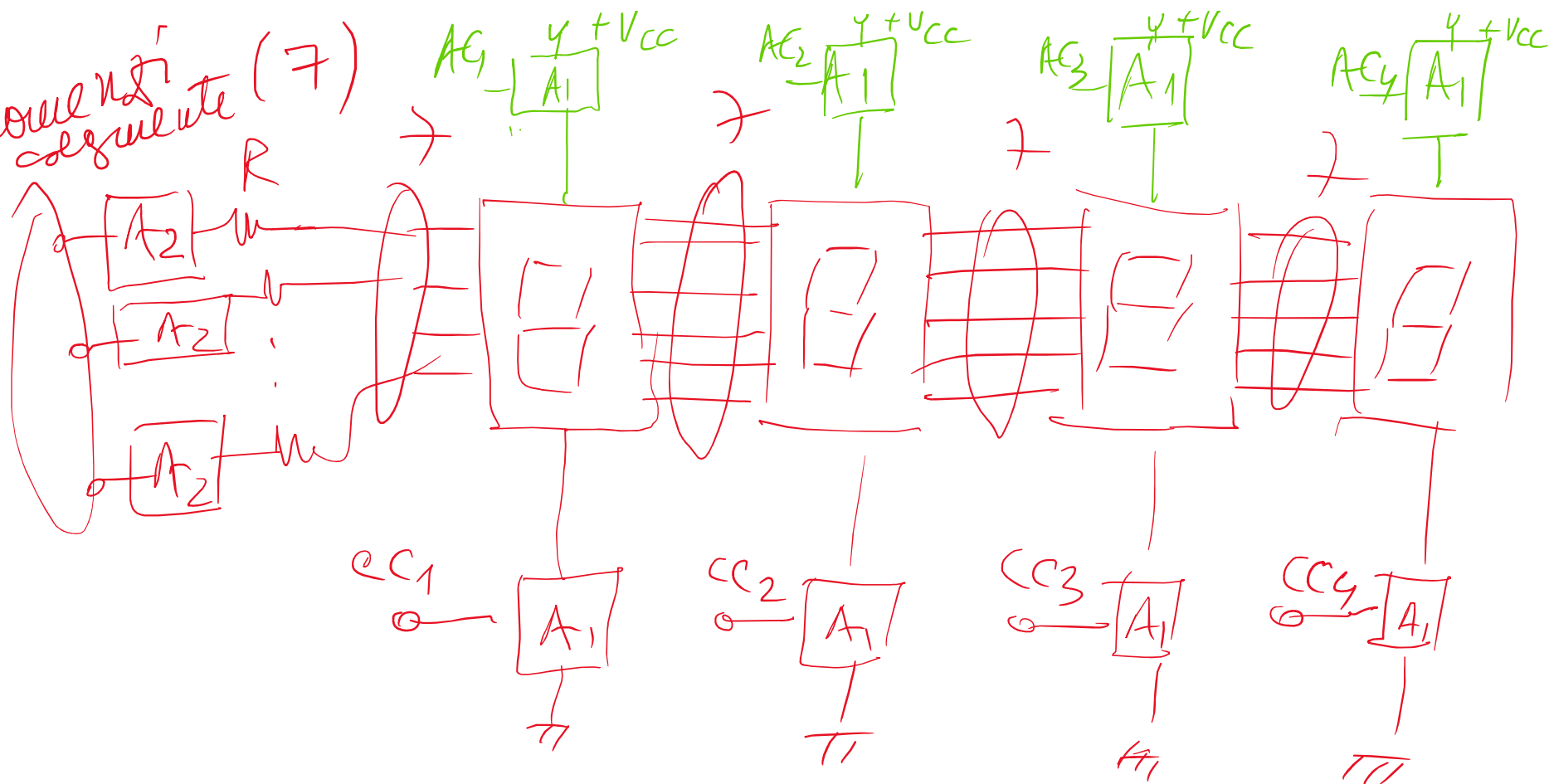
CC

$$V_{OL} \leq 0,4V$$

$$V_{OH} \geq 2,4V$$

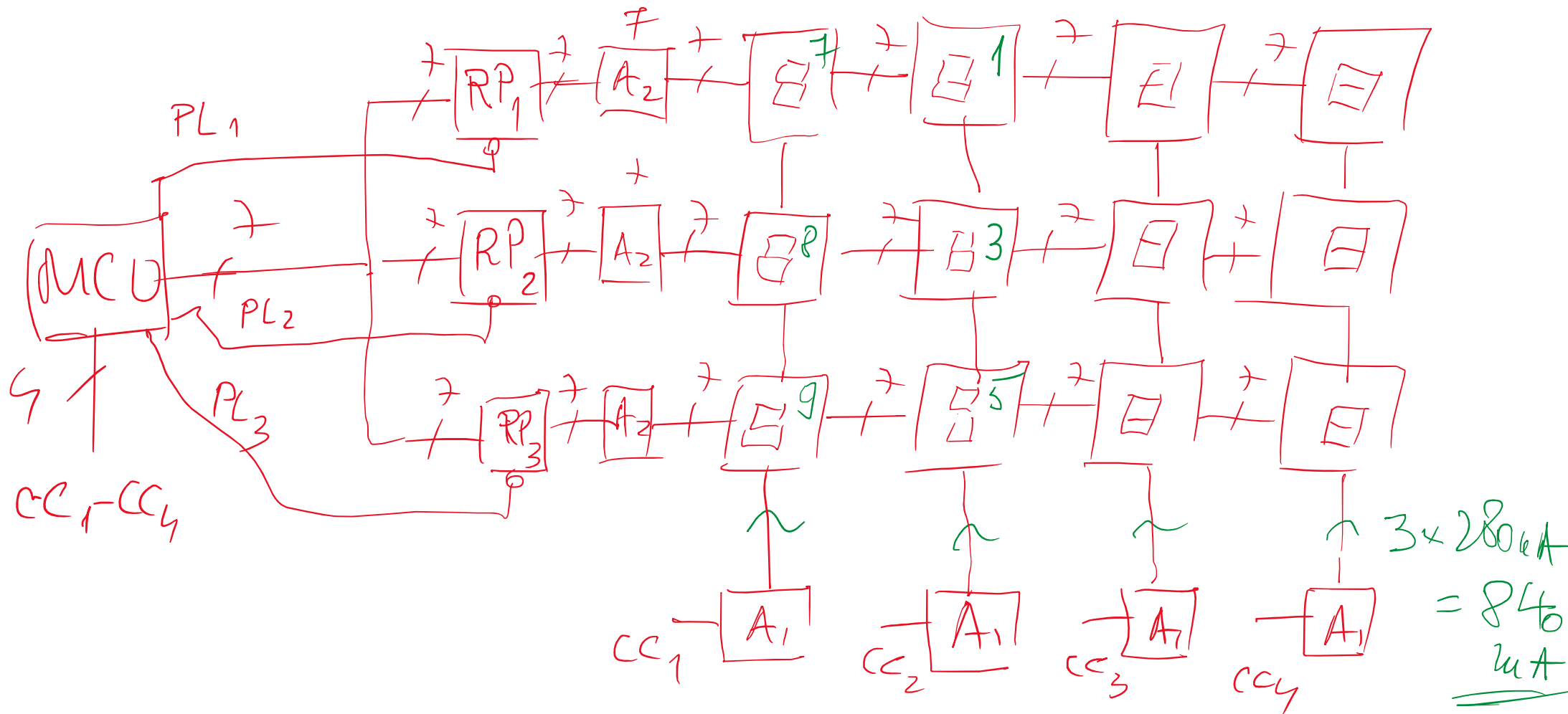


could not
converge (7)



AC

PL-parallel load

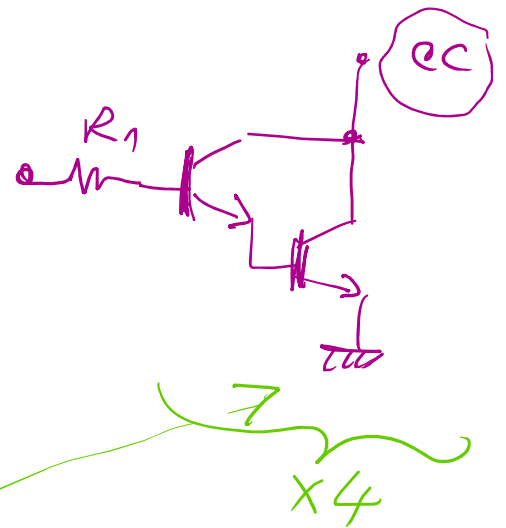
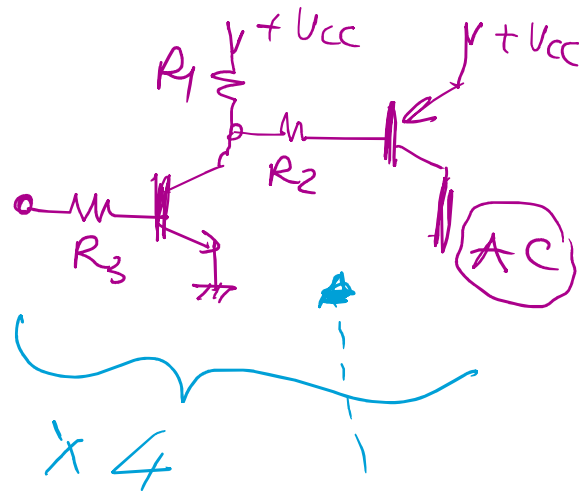
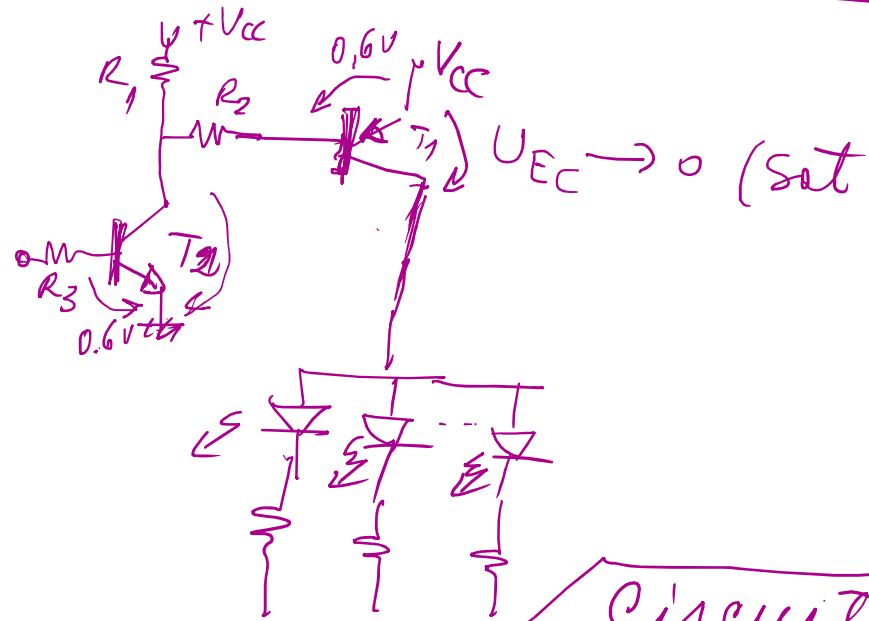


SURSE

- 1) Căutare celule LED 7seg (www.farnell.com)
- 3) Căutare tranzistoare (—||—) pentru A_1 și A_2
- 2) Scheme principale de multiplexare pentru matricea de celule 7seg din tablă (calcul curent)
- 4) Proiectarea A_1 și A_2 (valori de rezistențe + alegere tranzistor)

ANOD COMUN

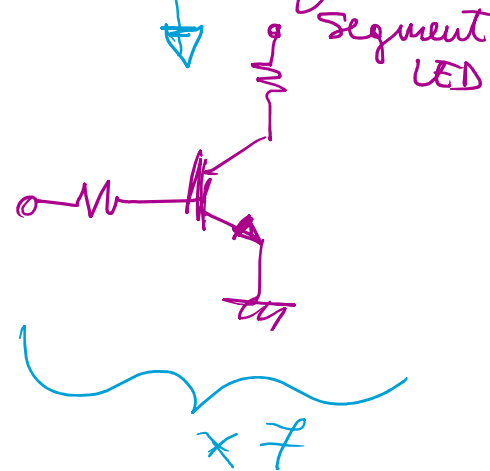
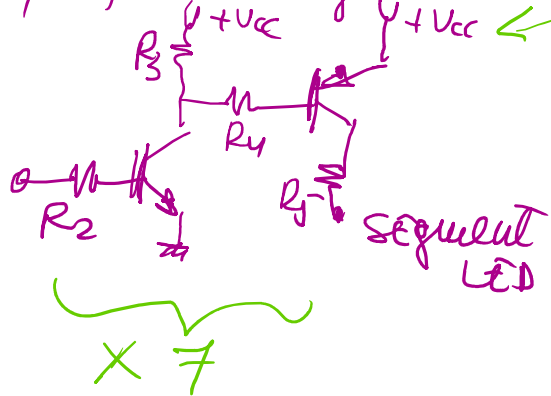
CA2

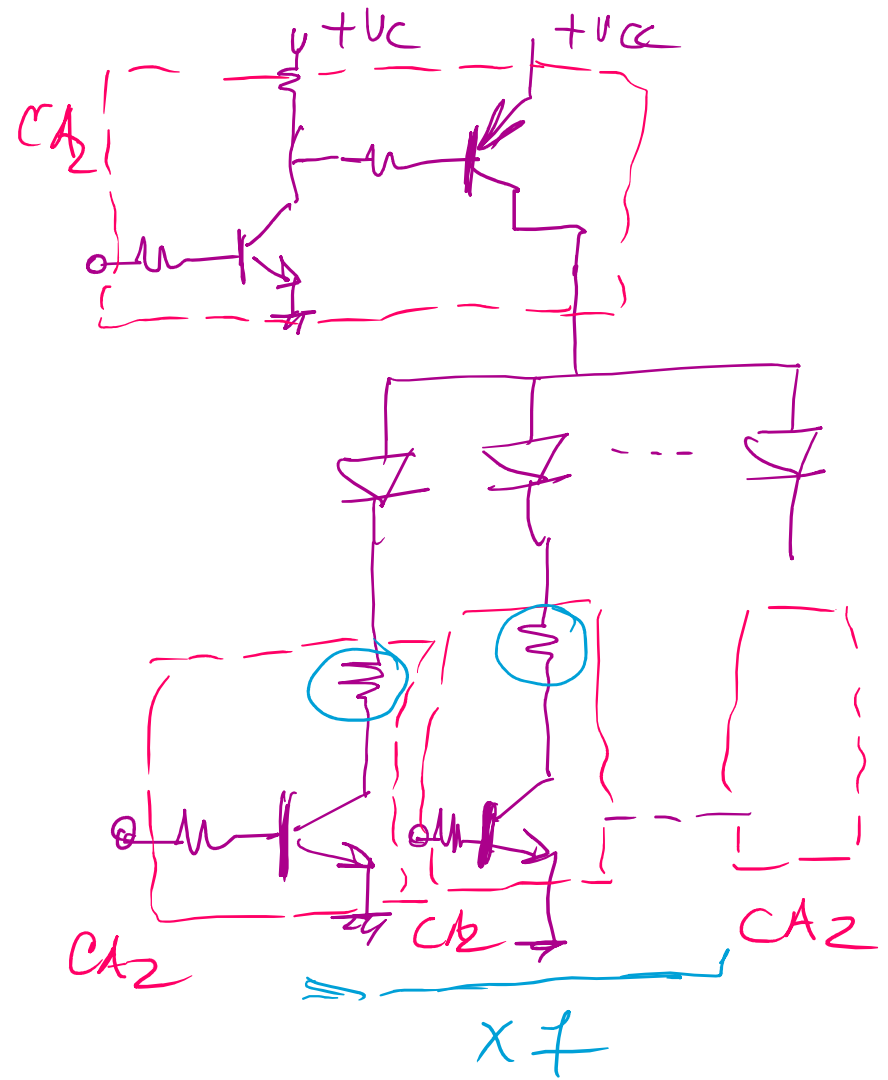
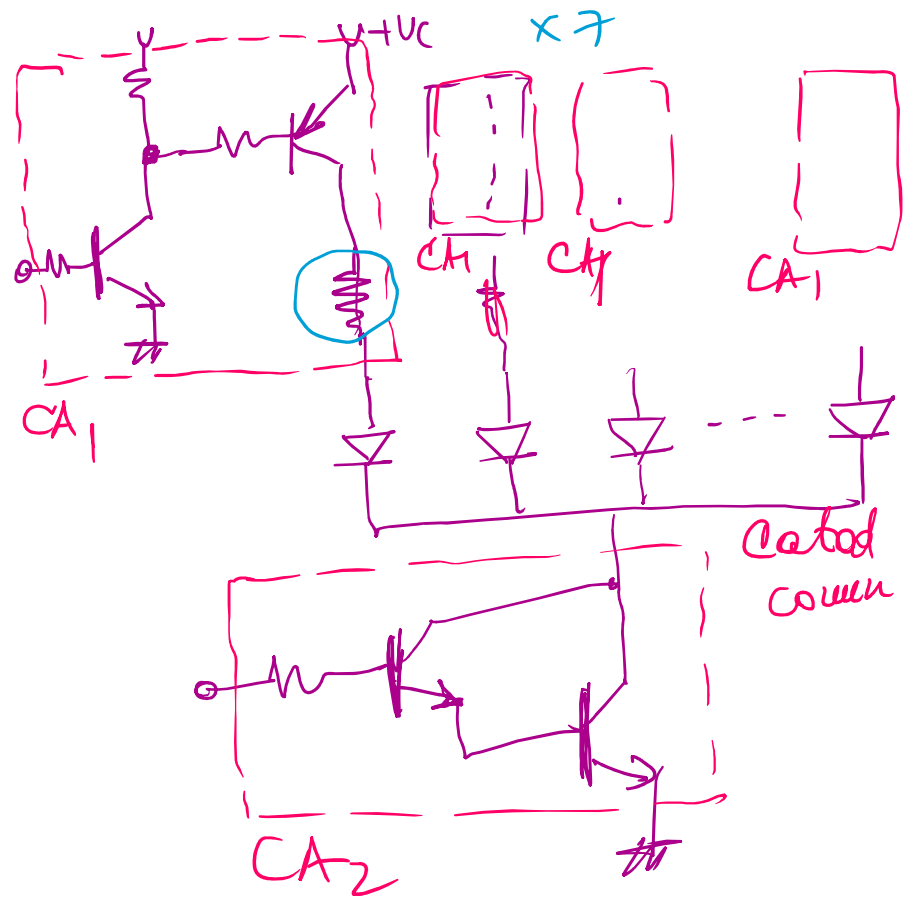


Circuit amplifcare curent pe segmente (CA1)

CATOD COMUN

pe fiecare segment



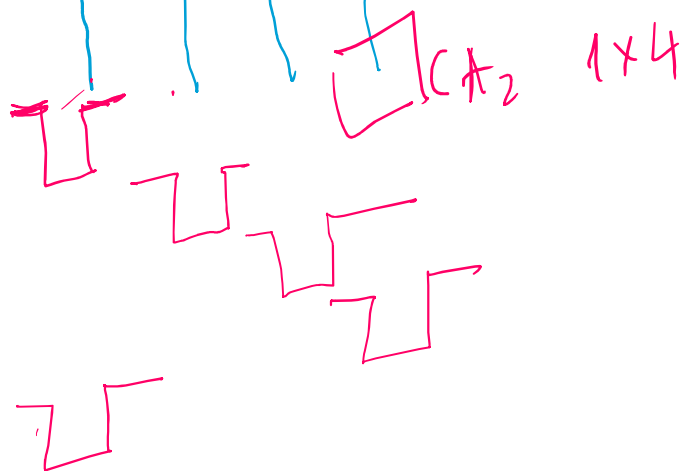
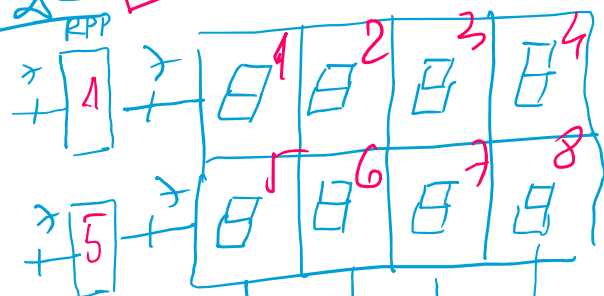


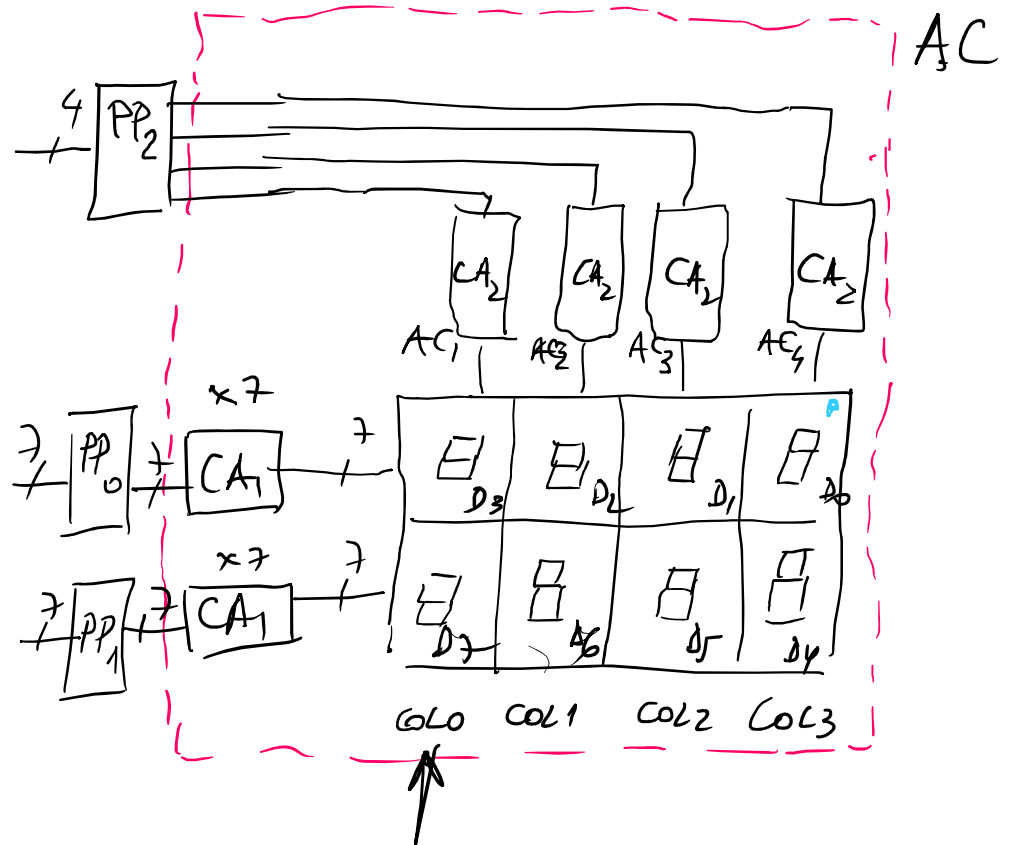
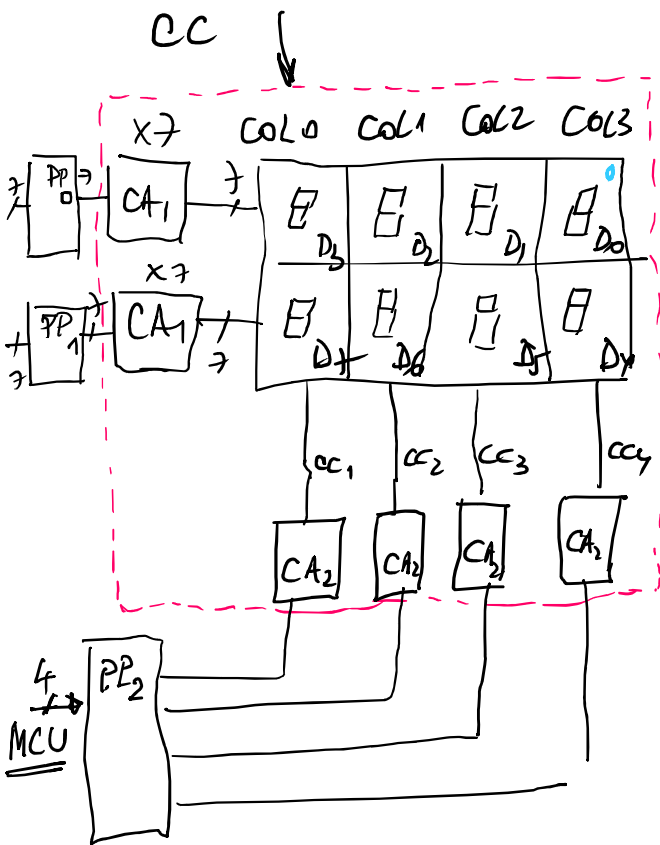
Disponere celule afroaj

caz 1



caz 2 $\square CA_1 \rightarrow 2 \times 4$





END 4 mar 2026

D ₃	D ₂	D ₁	D ₀
D ₇	D ₆	D ₅	D ₄

(comando electrice)

D ₃	D ₄	D ₅	D ₆	D ₇	D ₈	D ₉	D ₀
----------------	----------------	----------------	----------------	----------------	----------------	----------------	----------------

t_0 1 3 5 9 7 8 4 0
 t_1 3 5 9 7 8 4 0 1
 t_2 5 9 7 8 4 0 1 3

RAM

CONV.
 BIN
 7 SG

D ₀
D ₁
D ₂
D ₃
D ₄
D ₅
D ₆
D ₇
DBIN ₀
DBIN ₁
DBIN ₂
DBIN ₃
DBIN ₄
DBIN ₅
DBIN ₆
DBIN ₇

⇒ programme afisare

⇒ afisare conect

1

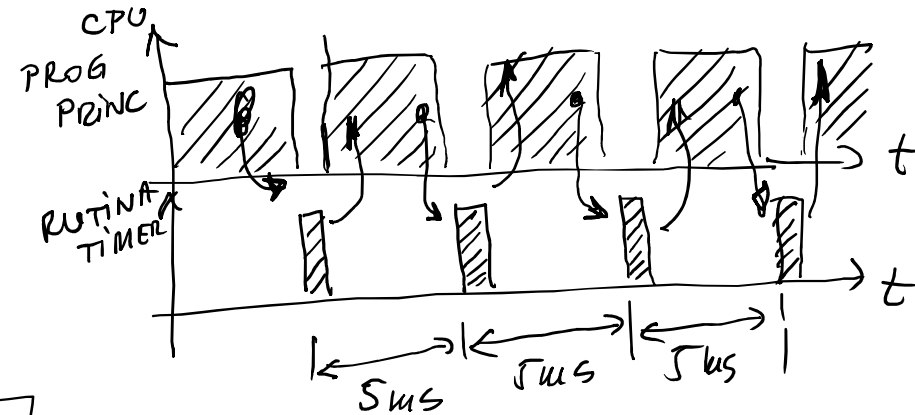
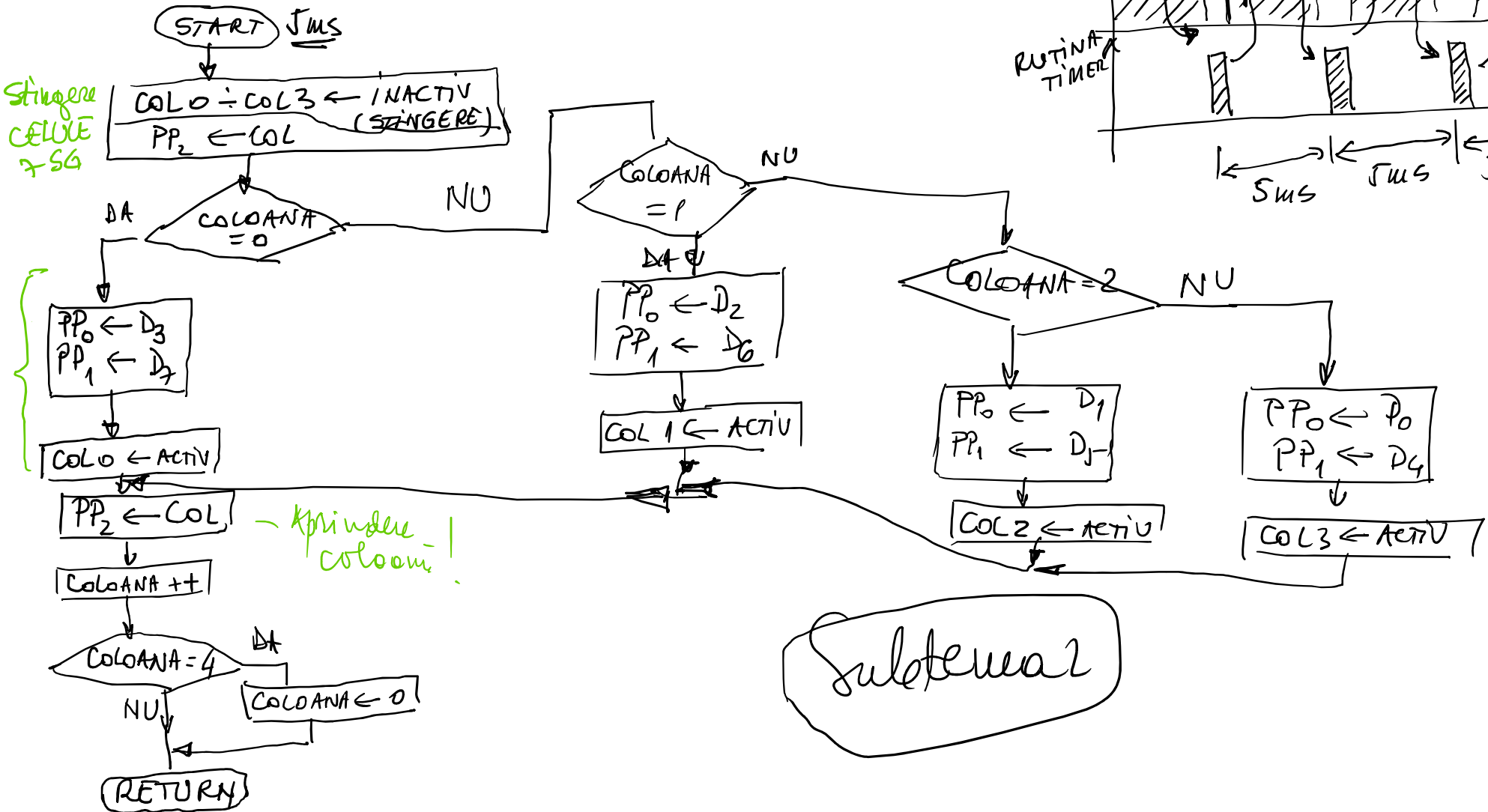
00000001
00000010
00000011
00000100

INC
DIGIT + 1

COL → COL 0, COL 1, COL 2, COL 3

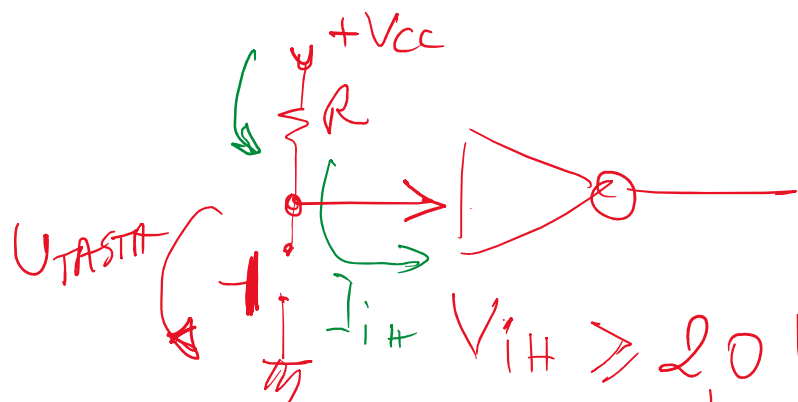
RUTINA TRATARE
INTRERUPERE TIMER
(SEMNALIZARE TRECERE 5ms)

stingere
celule
7-54



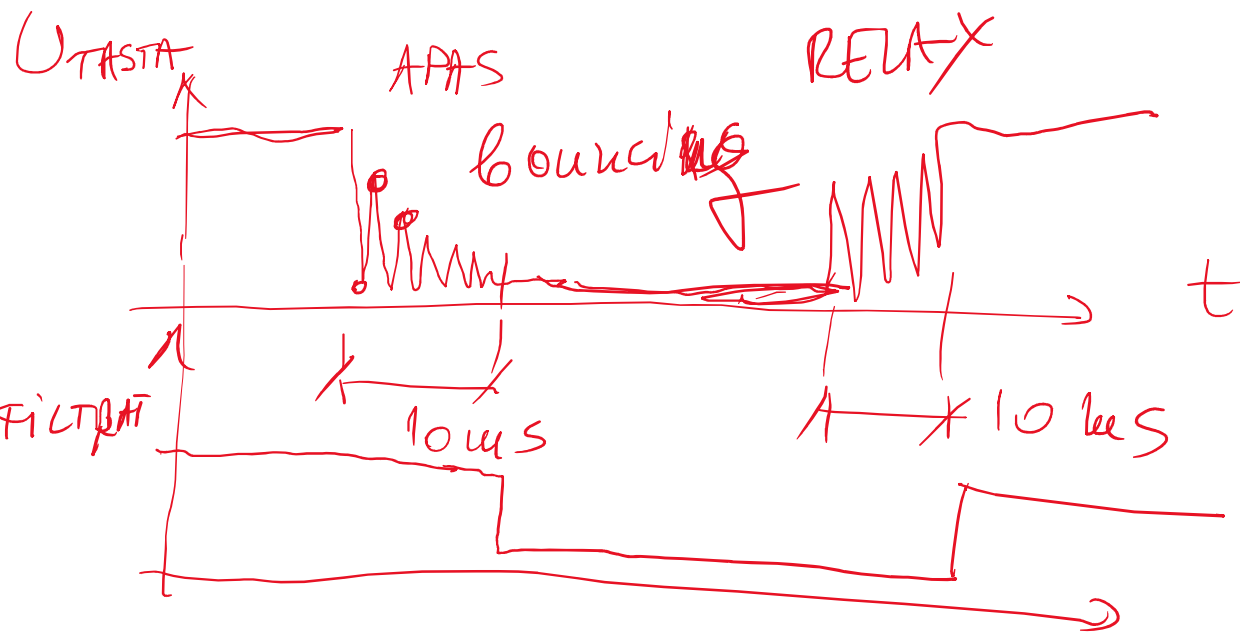
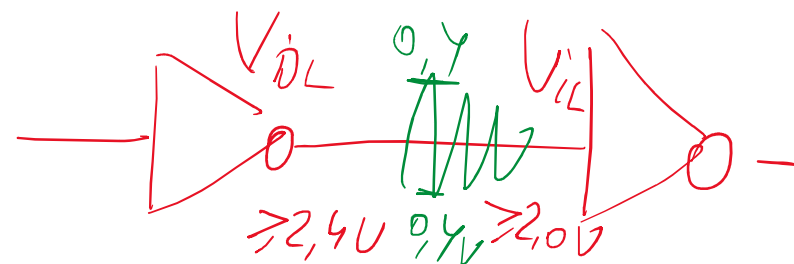
Nume variabila	Semnificatie	Valoarea de initializare	Mod de reprezentare	Adresa de stocare in memorie RAM
D0-D7	Imagini SW format 7 sg pentru celulele LED	7f(h) – CC; 00(h) – AC (segmente aprinse)	octet (8) – 7 biti utili / octet	30(h) – 37(h)
DBIN0-DBIN7	Imagini SW format binar (BCD) pentru celulele LED	08(h)	octet (8) – 4 biti utili / octet	38(h) – 3f(h)
COL0-COL3 (COL)	Imagini SW pentru comenzi pe coloanele comune	01(h) – AC; 0e(h) - CC	4 biti – accesibili individual	20(h) – zona adresabila pe Bit (00h, 01h, 02, 03h)
COLOANA	Indica ce coloana se actioneaza	03 – 00 (4 valori)	Octet	40(h)

END 11 mar 2026

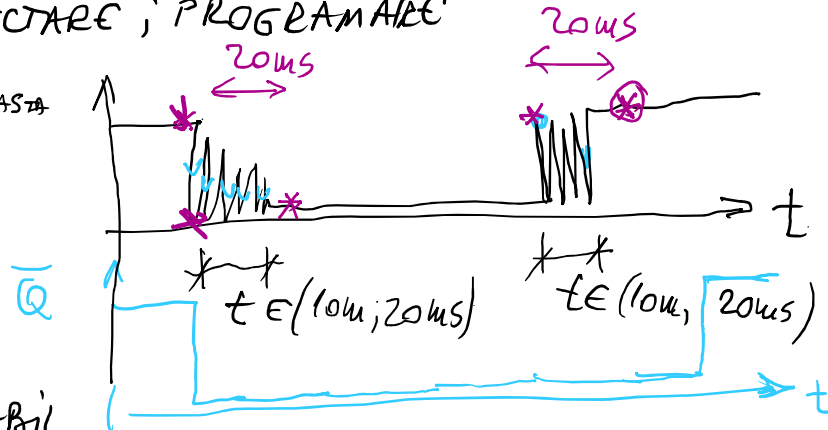
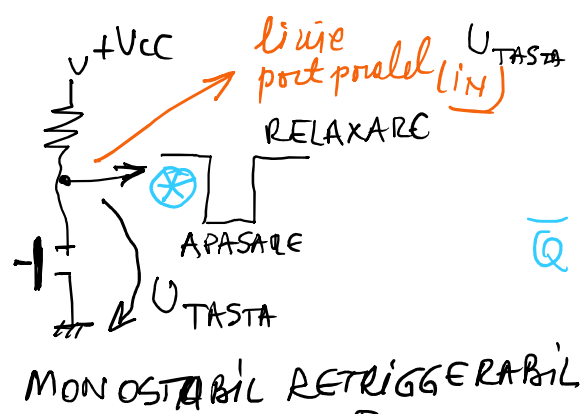


$$V_{iH} \geq 2,0V - V_{OH} \geq 2,4V$$

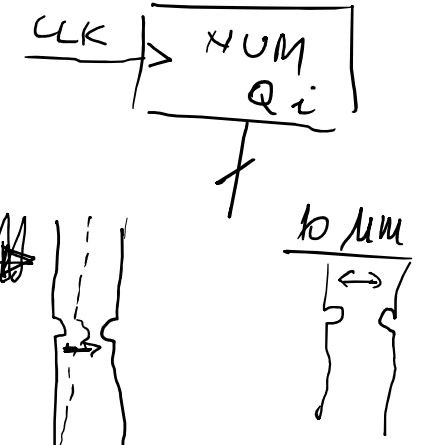
$$V_{iL} \leq 0,8V - V_{OL} \leq 0,4V$$



TASTATURA; CONECTARE; PROGRAMARE

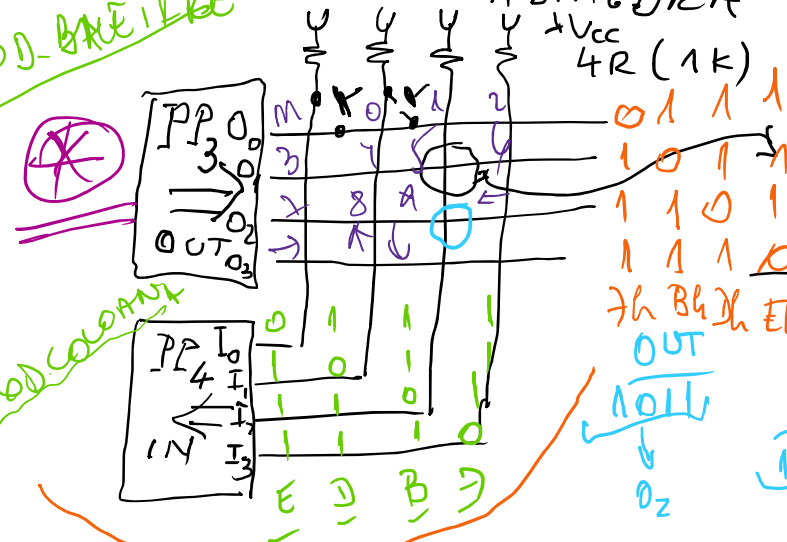


bounce debouncing



COMANDA MULTIPLEXATA TASTATURA

COD-BINEIERE



SUBTEMA 3 (PARTE HW TASTATURA)

20K - 1M

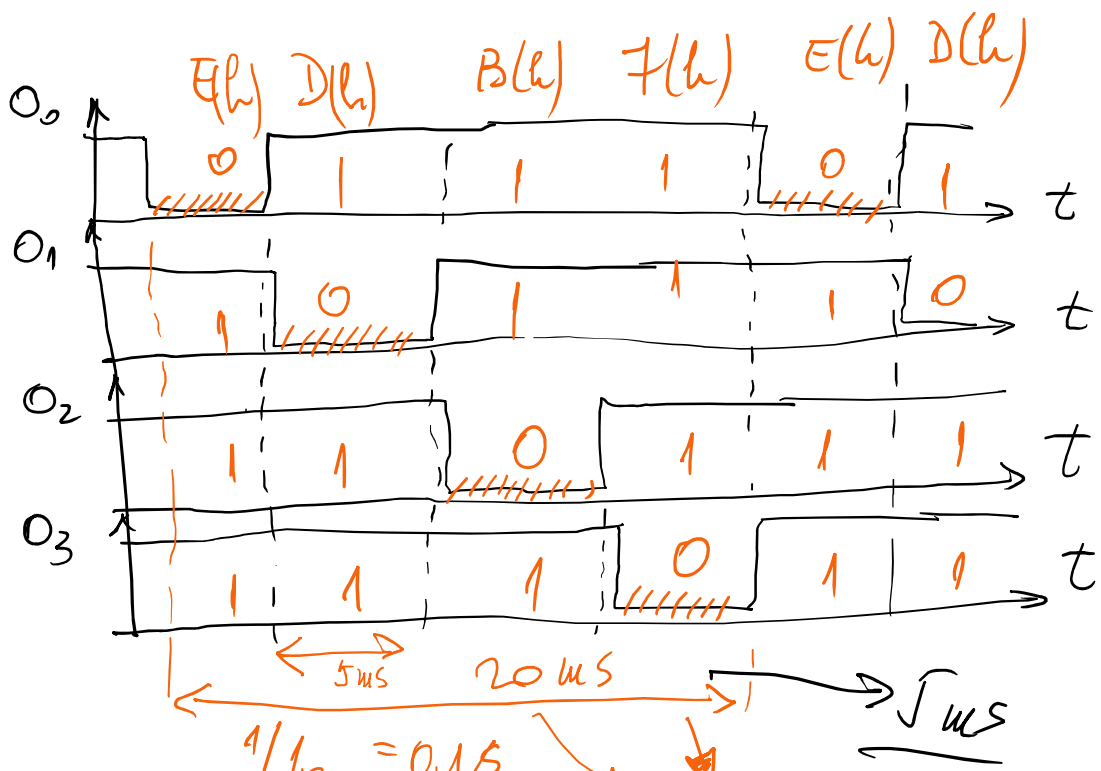
5V - 250 μV

folosire optima
linii PP

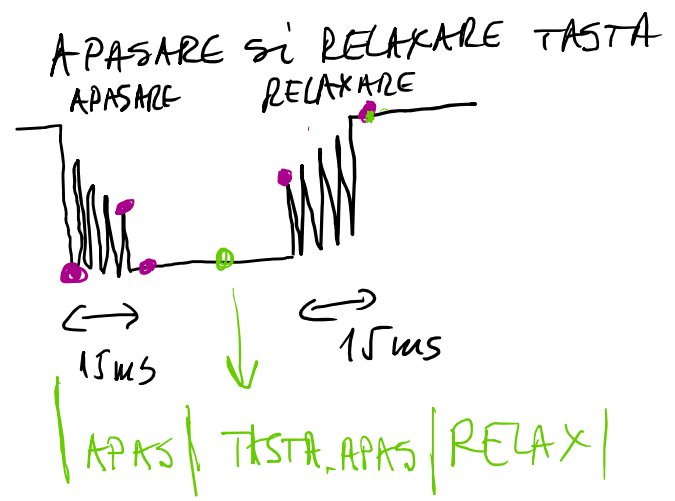
14 TASTE

2 x 7 = 9#

4 x 4 = 8#

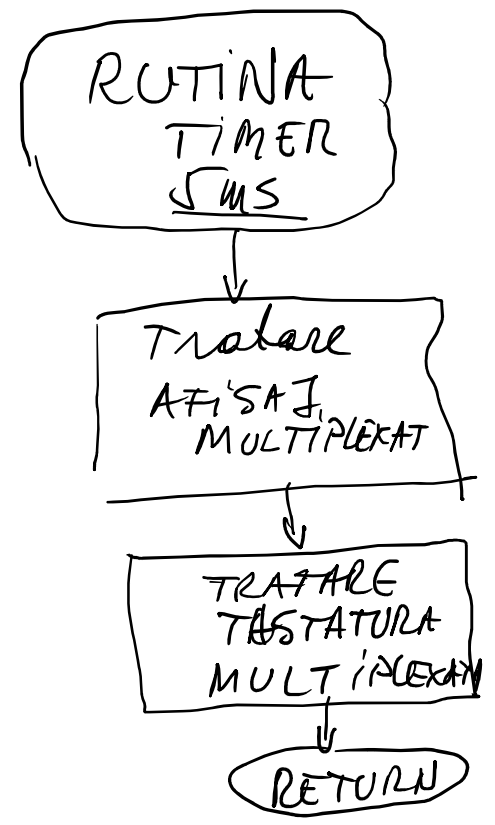


$\frac{1}{10} = 0,15$
 [100ms]
 50Hz
 15ms



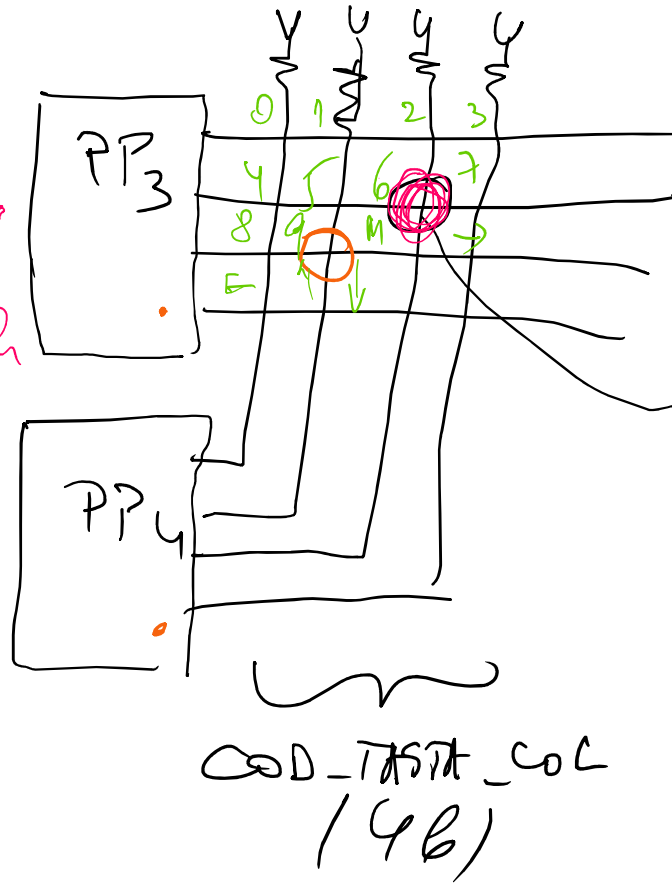
Proces de baleiere a tastaturii -

$E \rightarrow D \rightarrow B \rightarrow 7 \rightarrow E$
 - - - - -





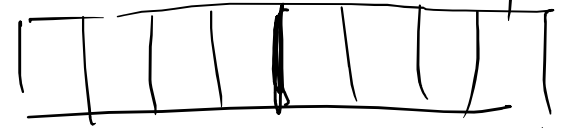
cod baleiaj
 $\in \{07h, 08h, 09h, 0Eh\}$



bicus

07h 08h
 0 1 1
 1 0 1
 1 1 0
 1 1 1

identificator tastei apasat



COD-BALEIA (4b) COD-TASTA-COL (4b)

1 0 1 1 1 1 0 1
 B D

SEMAFOR-TASTA - indica taste apasate corect
 COD-BALEIA } taste apasate
 COD-TASTA-COL }
 => progr. principal

0
 B Dh

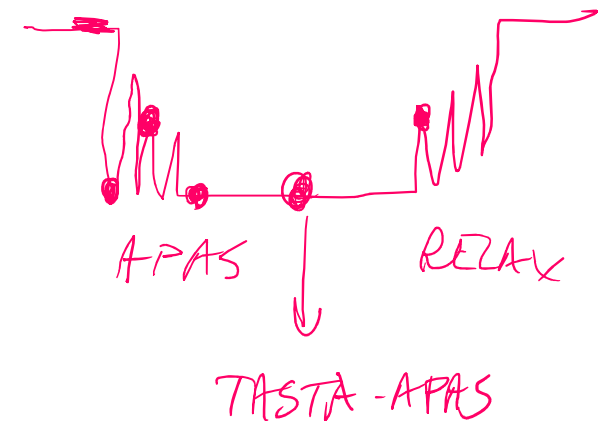
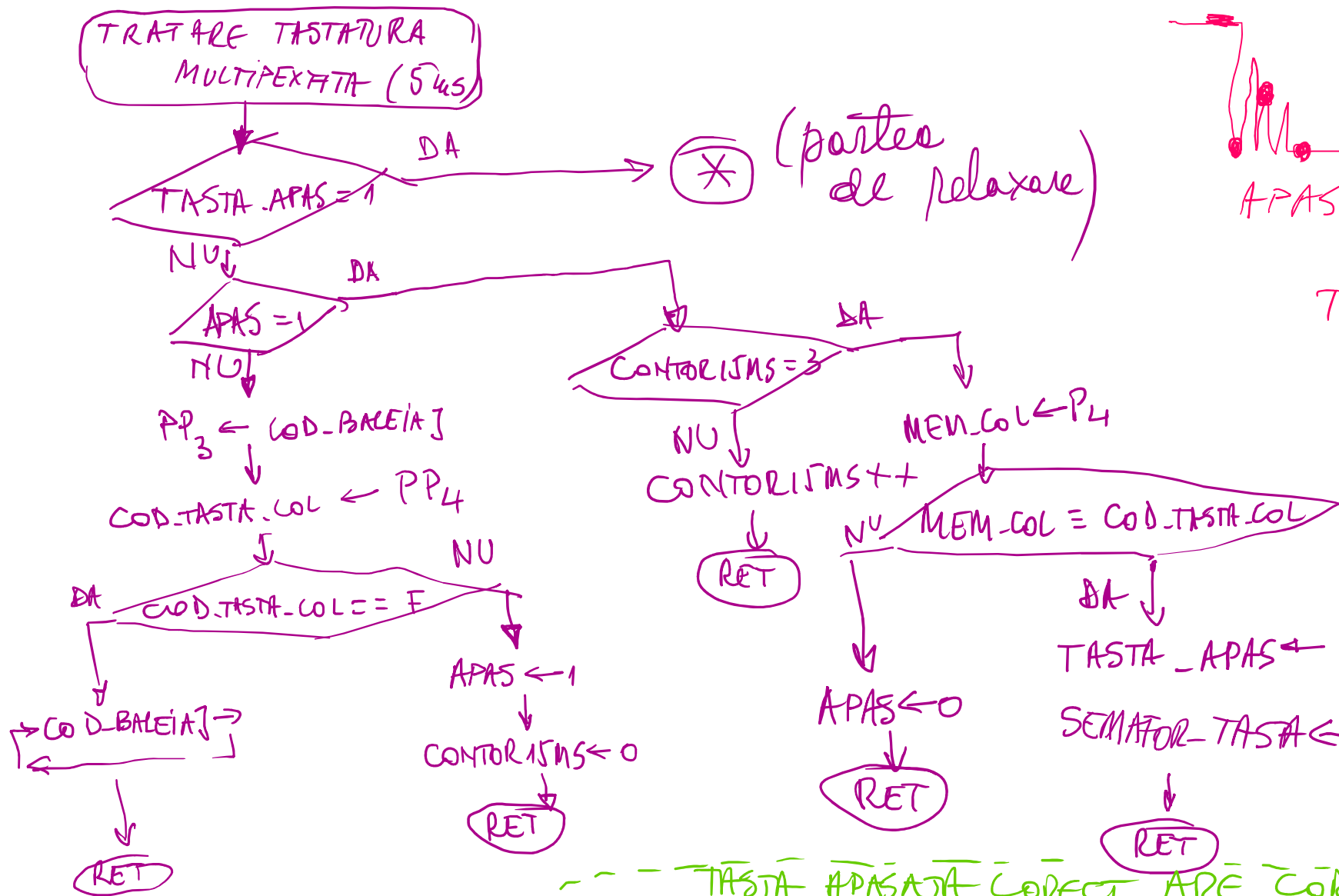
⊗ La lucru cu o tastatură este necesar:

- A - identificarea momentului apăsării unei taste
- B - realizarea unei temporizări (15 ms) pentru evitarea bouncing-ului
- C - verificarea stării de apăsare a tastei respective (0)
 - identificarea momentului relaxării unei taste
 - realizarea unei temporizări (15 ms) pentru evitarea bouncing-ului
 - verificarea stării de relaxare a tastei (1)

⊗ Prelucrarea unei taste apăsate de către o altă componentă este necesară:

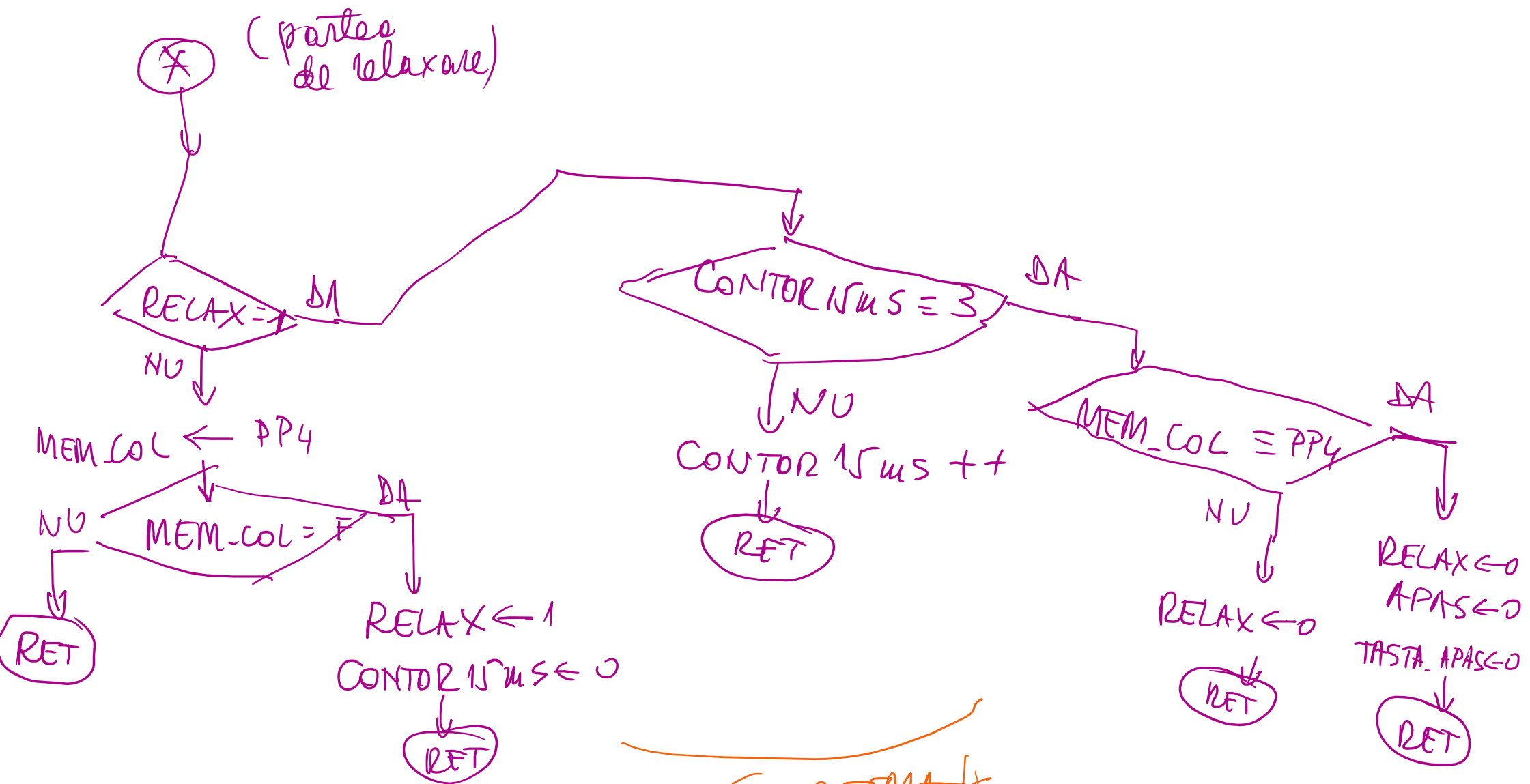
- semnalizarea unei taste corect apăsate (prin A, B, C)
(variabila SEMAFOR-TASTA)
- codul de identificare a tastei apăsate (variabilele obținute prin concatenarea cod de baleiere-pe linii și cod citit coloană)
COD - BALEIAJ COD - TASTA - COL

END 18 mar 2026



--- TASTA APASATA CORECT ARE CODUL DE IDENT.
FORMAT AIN COD.BALEIA SI COD.TASTA.COL!

INALTA
CATRE
PROG. PRINC
APASARE
CORECTA TASTA!



SUBTEMA 4

(SW - TASTATURA MUX.)

Manual de utilizare al tabelor

Subtema 5

① Tabelor are 2
moduri de funcționare
1) mod 0 - operare
normale

2) mod 1 - programare

Modul se scrie prin
apăsarea tastei M

② În modul operare
normale

- textul afișat se deplasează
cațre stânga cea viteză
de deplasare indicată în funcție

- supraveghere taste M

și creșterea mod -
= nici o altă tastă nu
acionează în modul 0 !

③ Modul programare
= textul afișat rămâne fix
= primul digit (dreapta) va clipi
cu timpul indicat în funcție

= supraveghere taste apăsate

0, ..., 9, $\leftarrow \uparrow \rightarrow$, M

= la tastele 0 ÷ 9 → se programează
digitul
cu valoarea

= cu tastele $\uparrow$ - incrementare
 $\downarrow$ - decrementare

= cu tastele $\leftarrow \Rightarrow$ aleg
următorul digit care se
programează $\leftarrow$



END 25 mar 2026